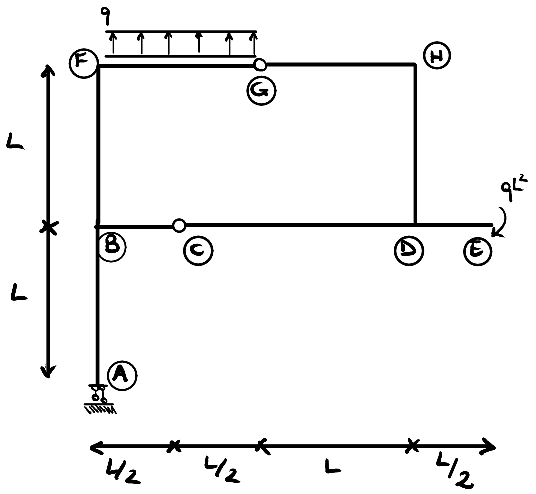
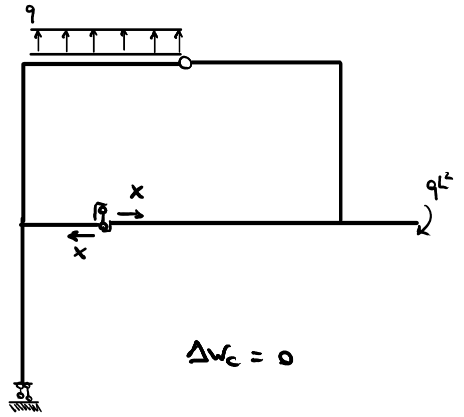
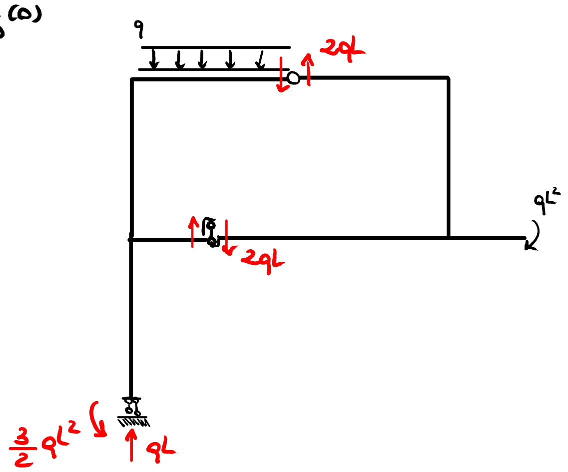


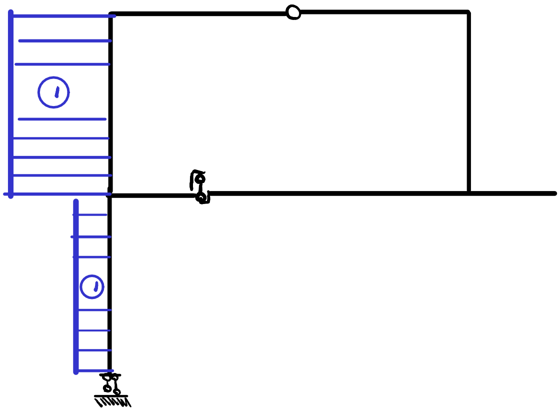
$$\left.\begin{aligned} t &= 2 \\ s &= 6 \\ \ell &= 1 \end{aligned}\right\} i = 1$$

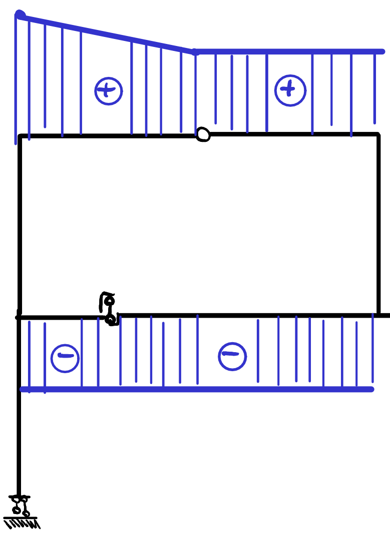


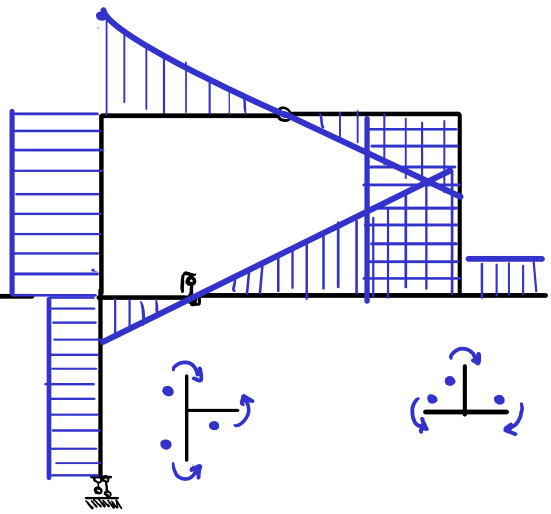
$\Delta w_c = 0$

$S^{(0)}$









$N_{AB}^{(0)} = -qL$

$N_{BF}^{(0)} = -3qL$

$N_{BC}^{(0)} = 0$

$N_{FG}^{(0)} = 0$

$N_{GH}^{(0)} = 0$

$N_{CD}^{(0)} = 0$

$N_{HD}^{(0)} = 2qL$

$N_{DE}^{(0)} = 0$

$T_{AB}^{(0)} = 0$

$T_{BF}^{(0)} = 0$

$T_{BC}^{(0)} = -2qL$

$T_{FG}^{(0)} = 2qL + q(L-z)$

$T_{GH}^{(0)} = 2qL$

$T_{CD}^{(0)} = -2qL$

$T_{HD}^{(0)} = 0$

$T_{DE}^{(0)} = 0$

$M_{AB}^{(0)} = -\frac{3}{2}qL^2$

$M_{BF}^{(0)} = -\frac{3}{2}qL^2 - 2qL \cdot \frac{L}{2}$

$M_{BC}^{(0)} = 2qL\left(\frac{L}{2} - z\right)$

$M_{FG}^{(0)} = -2qL(L-z) - q\frac{(L-z)^2}{2}$

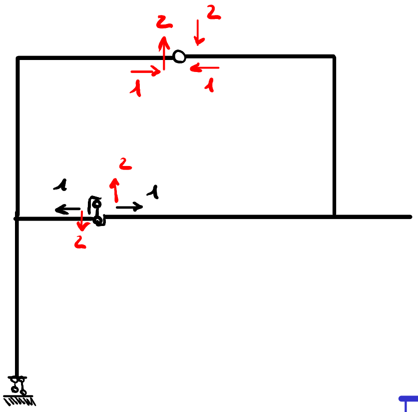
$M_{GH}^{(0)} = 2qLz$

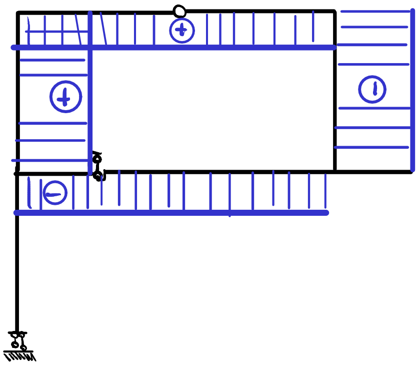
$M_{CD}^{(0)} = -2qLz$

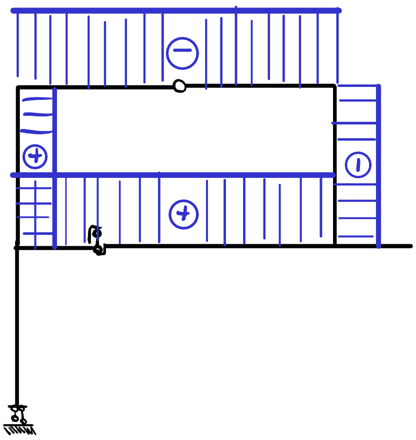
$M_{HD}^{(0)} = 2qL^2$

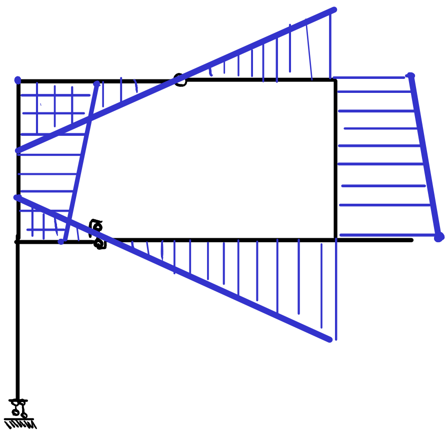
$M_{DE}^{(0)} = -qL^2$

$S^{(1)}$









$N_{AB}^{(1)} = 0$
$N_{BF}^{(1)} = 2$
$N_{BC}^{(1)} = -1$
$N_{FG}^{(1)} = 1$
$N_{GH}^{(1)} = 1$
$N_{CD}^{(1)} = -1$
$N_{HD}^{(1)} = -2$
$N_{DE}^{(1)} = 0$

$T_{AB}^{(1)} = 0$
$T_{BF}^{(1)} = 1$
$T_{BC}^{(1)} = 2$
$T_{FG}^{(1)} = -2$
$T_{GH}^{(1)} = -2$
$T_{CD}^{(1)} = 2$
$T_{HD}^{(1)} = -1$
$T_{DE}^{(1)} = 0$

$M_{AB}^{(1)} = 0$
$M_{BF}^{(1)} = z + 2\frac{L}{2}$
$M_{BC}^{(1)} = -2\left(\frac{L}{2} - z\right)$
$M_{FG}^{(1)} = 2(L - z)$
$M_{GH}^{(1)} = -2z$
$M_{CD}^{(1)} = 2z$
$M_{HD}^{(1)} = -2L - z$
$M_{DE}^{(1)} = 0$

$$\mathcal{L}_{ve} = 1 \cdot \Delta w_c = 0$$

$$\mathcal{L}_{vi} = \sum_{i,j} \int_i^j \frac{M_{ij}^{(1)}\left(M_{ij}^{(0)} + x M_{ij}^{(1)}\right)}{EI}\, dz$$

Ipotesi: tutti i tratti hanno $EI \to +\infty$ tranne AB e BC

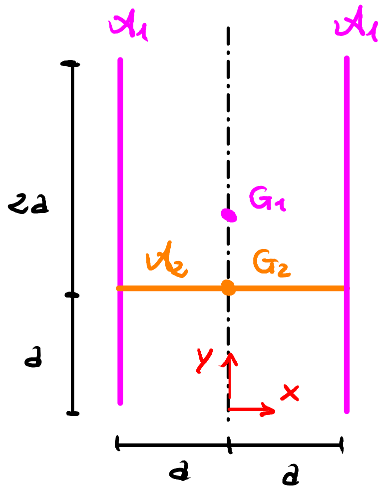
$$\Rightarrow \mathcal{L}_{vi} = \int_0^L \frac{\cancel{M_{AB}^{(1)}}\left(M_{AB}^{(0)} + x M_{AB}^{(1)}\right)}{EI}\, dz + \int_0^{L/2} \frac{M_{BC}^{(1)}\left(M_{BC}^{(0)} + x M_{BC}^{(1)}\right)}{EI}\, dz =$$

$$= \int_0^{L/2} \frac{-2\left(\frac{L}{2} - z\right)\left(2qL\left(\frac{L}{2} - z\right) - 2X\left(\frac{L}{2} - z\right)\right)}{EI}\, dz = \frac{(X - qL)L^3}{6EI}$$

$$\mathcal{L}_{ve} = \mathcal{L}_{vi} \Rightarrow X = qL$$

## Sezione retta

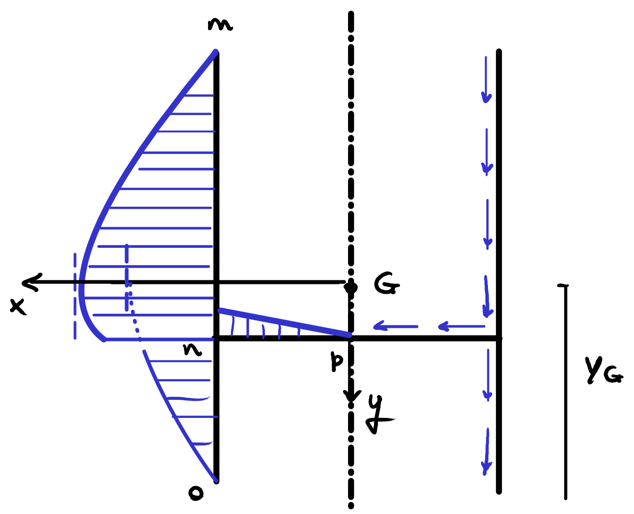
Rispetto a $\{x, y\}$
$$\begin{cases} X_{G1} = 0, & Y_{G1} = \frac{3}{2}a, & A_1 = 2(3a)s = 6as \\ X_{G2} = 0 & Y_{G2} = a, & A_2 = 2as \end{cases}$$




$$x_G = 0$$

$$y_G = \frac{\sum_i y_{Gi} A_i}{A} = \frac{\frac{3}{2}a \cdot 6as + a \cdot 2as}{8as} = \frac{11}{8}a$$

## Tensioni tangenziali alla Jouravsky




$$\tau_{zt} = -\frac{T_y}{s I_x} S_x^*$$

$$S^*_{x\,mn} = -ts\left(3a - y_G - \frac{t}{2}\right) = -ts\left(\frac{13}{8}a - \frac{t}{2}\right)$$

$$S^*_{x\,on} = ts\left(y_G - \frac{t}{2}\right)$$

$$S^*_{x\,np} = S^*_{x\,mo} + ts\,(y_G - a) = -\frac{3}{8}a^2 s + \frac{3}{8}a t s$$

$$S^*_{x\,np} = S^*_{x\,mo} + st\,(y_G - a)$$

Tre modi equivalenti per calcolare $S^*_{x\,mo}$ :

$$S^*_{x\,mo} = S^*_{x\,mn}(t = 2a) + S^*_{x\,on}(t = a) = -2as\left(\frac{13}{8}a - \frac{2a}{2}\right) + as\left(\frac{11}{8}a - \frac{a}{2}\right) =$$
$$= -2as\left(\frac{5}{8}a\right) + as\left(\frac{7}{8}a\right) = \left(-\frac{5}{4} + \frac{7}{8}\right)a^2 s = -\frac{3}{8}a^2 s$$

$$S^*_{x\,mo} = S^*_{x\,mn}(t = 3a) = -3as\left(\frac{13}{8}a - \frac{3a}{2}\right) = -3as\left(\frac{a}{8}\right) = -\frac{3}{8}a^2 s$$

$$S^*_{x\,mo} = S^*_{x\,on}(t = 3a) = 3as\left(\frac{11}{8}a - \frac{3a}{2}\right) = 3as\left(-\frac{a}{8}\right) = -\frac{3}{8}a^2 s$$